Shahjalal University of Science and Technology, Sylhet

Department of Electrical & Electronic Engineering

Course Code: EEE 0713 3237 Course Title: Power System II

Semester Final Examination (OBE): **Final Model Question**

SET 1

Practice paper for self-study, mapped to CO1–CO5. Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

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*[Marks · CO · BL = Bloom level · Src: **PYQ** previous-year, **TB** textbook example/exercise, **LN** lecture-note article, **EXT** external-examiner style. Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]*

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

		Marks	CO	BL	Src
1.	(a) What is transposition of a transmission line? Why is it used? *[Ref: Asked 2021, 2022, 2024 ×2]*	[2]	CO1	C2	**PYQ**
	(b) Show why unsymmetrical spacing makes the phase inductances unequal, and derive the inductance per phase of a fully transposed three-phase line. *[Ref: Kothari & Nagrath, Ch-2, Art 2.8, p.82]*	[3]	CO1	C3	**LN**
	(c) The forward circuit of a single-phase line is composed of three solid wires of 2.5 mm radius and the return circuit of two wires of 5 mm radius, placed symmetrically as shown in Fig. 1. Find the inductance of each side of the line and of the complete line.	[3]	CO1	C3	**TB**

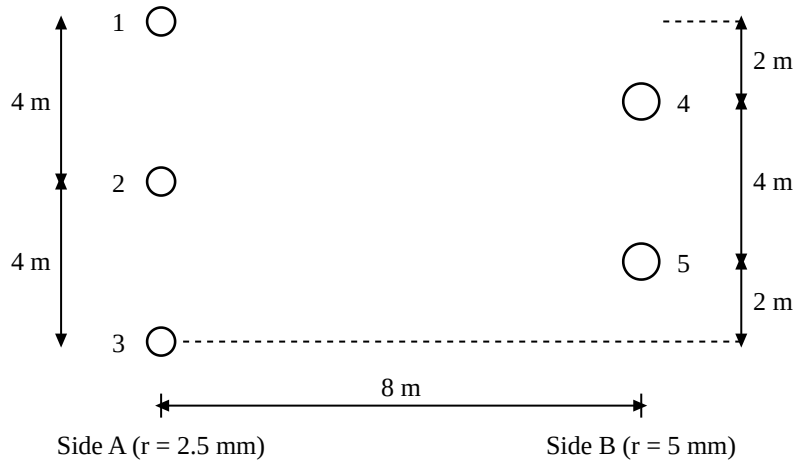


Fig. 1

[Ref: Kothari & Nagrath, Ch-2, Example 2.3, p.81]

	(d) A utility wants to reduce the reactance of a 132 kV line. It is considering (i) a larger phase spacing and (ii) a 2-conductor bundle per phase. Which option is better for reactance and for corona? Justify your choice.	[2]	CO5	C5	**EXT**

2.	(a)	Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing (fully transposed). *[Ref: Asked 2022, 2024 ×2, TT-1]*	[3]	CO1	C3	**PYQ**
	(b)	Find the capacitance to neutral per km of a three-phase line with conductors of 2 cm diameter placed at the corners of a triangle with sides 5 m, 6 m and 7 m. Also find the charging current per km if the line operates at 132 kV, 50 Hz. Assume full transposition. *[Ref: Kothari & Nagrath, Ch-3, Problem 3.8, p.111]*	[3]	CO1	C3	**TB**
	(c)	Calculate the capacitance to neutral per km of a single-phase line of No. 2 single-strand conductors (radius 0.328 cm) spaced 3 m apart and 7.5 m above the ground, (i) neglecting and (ii) considering the effect of earth. *[Ref: Kothari & Nagrath, Ch-3, Example 3.1, p.103]*	[2]	CO1	C3	**TB**
	(d)	A 220 kV, 50 Hz, 150 km line has a capacitance to neutral of 0.011 μF/km. Find the total charging MVAr. Why are shunt reactors connected to such lines at light load?	[2]	CO5	C4	**EXT**
3.	(a)	With a neat figure, describe the general construction of a 3-core underground cable. Why are belted cables limited to about 11 kV? *[Ref: V.K. Mehta, Ch-11, Art 11.1, 11.5, p.268]*	[3]	CO2	C2	**LN**
	(b)	A single-core cable has a conductor diameter of 1 cm and insulation thickness of 0.4 cm. If the specific resistance of insulation is 5×10^{14} Ω-cm, calculate the insulation resistance for a 2 km length. *[Ref: V.K. Mehta, Ch-11, Example 11.1, p.273]*	[2]	CO2	C3	**TB**
	(c)	A single-core cable for use on 11 kV, 50 Hz has a conductor area of 0.645 cm^2 and an internal sheath diameter of 2.18 cm. The permittivity of the dielectric is 3.5. Find (i) g_{max}, (ii) g_{min}, (iii) capacitance per km and (iv) charging current. *[Ref: V.K. Mehta, Ch-11, Example 11.9, p.279]*	[3]	CO2	C3	**PYQ**
	(d)	For 11 kV power distribution in a congested city area, would you choose overhead lines or underground cables? Justify on safety, environmental and cost grounds.	[2]	CO5	C5	**EXT**

OR

4.	(a)	What is capacitance grading? For three dielectrics, derive the expression for the potential difference between core and earthed sheath. *[Ref: V.K. Mehta, Ch-11, Art 11.12, p.281]*	[3]	CO2	C3	**LN**
	(b)	A single-core lead-sheathed cable is graded using three dielectrics of relative permittivity 5, 4 and 3. The conductor diameter is 2 cm and the overall diameter is 8 cm. If the three dielectrics are worked at the same maximum stress of 40 kV/cm, find the safe working voltage of the cable. What would it be for an ungraded cable of the same dimensions? *[Ref: V.K. Mehta, Ch-11, Example 11.12, p.282]*	[3]	CO2	C4	**TB**
	(c)	The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is 4.5×10^{14} Ω-cm, find the insulation thickness. *[Ref: V.K. Mehta, Ch-11, Example 11.2, p.274]*	[2]	CO2	C3	**PYQ**
	(d)	A supplier offers a graded and an ungraded 66 kV cable of the same overall size. Which would you buy and why? Why is intersheath grading rarely used in practice?	[2]	CO5	C5	**EXT**

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL	Src
5.	(a)	Draw the equivalent circuit of a 3-disc suspension string and derive V_2 and V_3 in terms of V_1 and K. Define string efficiency. *[Ref: V.K. Mehta, Ch-8, Art 8.7, p.168]*	[3]	CO2	C3	**LN**
	(b)	Each conductor of a 3-phase high-voltage line is suspended by a string of 4 suspension-type disc insulators. If the potential difference across the second unit from the top is 13.2 kV and across the third from the top is 18 kV, determine the voltage between conductors. *[Ref: V.K. Mehta, Ch-8, Example 8.8, p.176]*	[3]	CO2	C4	**TB**
	(c)	Each line of a 3-phase system is suspended by a string of 3 identical insulators of self-capacitance C. The shunt capacitance of the connecting metal work of each insulator is 0.2 C to earth and 0.1 C to line. Calculate the string efficiency if a guard ring increases the capacitance to the line of the metal work of the lowest insulator to 0.3 C. *[Ref: V.K. Mehta, Ch-8, Example 8.11, p.179]*	[2]	CO2	C4	**TB**
	(d)	On a coastal 132 kV line the disc nearest the conductor fails much more often than the others. Analyse the reason and suggest two remedies.	[2]	CO4	C4	**EXT**
6.	(a)	Define critical disruptive voltage. Write down its significance. *[Ref: Asked 2022, 2024 ×2]*	[2]	CO2	C1	**PYQ**
	(b)	A 3-phase line has conductors 2 cm in diameter spaced equilaterally 1 m apart. If the dielectric strength of air is 30 kV (max) per cm, find the disruptive critical voltage for the line. Take $\delta = 0.952$ and $m_0 = 0.9$. *[Ref: V.K. Mehta, Ch-8, Example 8.13, p.184]*	[2]	CO2	C3	**TB**
	(c)	A 3-phase, 220 kV, 50 Hz transmission line consists of 1.5 cm radius conductors spaced 2 m apart in an equilateral triangle. If the temperature is 40 °C and the atmospheric pressure is 76 cm, calculate the corona loss per km of the line. Take $m_0 = 0.85$. *[Ref: V.K. Mehta, Ch-8, Example 8.15, p.185]*	[3]	CO4	C3	**TB**
	(d)	Explain the hazards of corona to nearby communication lines and the environment. How does a designer avoid corona on a 400 kV line?	[3]	CO4	C5	**EXT**

OR

7.	(a)	Classify the faults in underground cables. How is a Megger used to identify each type? *[Ref: V.K. Mehta, Ch-11, Art 11.20, p.293]*	[2]	CO4	C2	**LN**
	(b)	With a neat diagram, derive the expression for the fault distance in the Murray loop test (earth fault). *[Ref: V.K. Mehta, Ch-11, Art 11.22, p.293]*	[2]	CO4	C3	**LN**
	(c)	In a test by Murray loop for a ground fault on 500 m of cable having a resistance of 1.6 Ω/km, the faulty cable is looped with a sound cable of the same length and area of cross-section. If the ratio of the other two arms of the testing network at balance is 3 : 1, find the distance of the fault from the testing end. *[Ref: V.K. Mehta, Ch-11, Example 11.22, p.297]*	[2]	CO4	C3	**TB**
	(d)	A Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 Ω. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 Ω and the fixed resistors are equal. Calculate the distance of the fault from the test end. *[Ref: V.K. Mehta, Ch-11, Example 11.24, p.297]*	[2]	CO4	C3	**PYQ**
	(e)	An 11 kV cable under a busy road has failed. Describe, step by step, how you would identify the fault type and locate it before digging. Why does a high fault resistance reduce the accuracy of a loop test?	[2]	CO4	C4	**EXT**
8.	(a)	Derive the swing equation for a synchronous machine. *[Ref: Asked 2021, 2022]*	[4]	CO4	C3	**PYQ**
	(b)	A 50 Hz, four-pole turbogenerator rated 100 MVA, 11 kV has an inertia constant of 8.0 MJ/MVA. (i) Find the stored energy in the rotor at synchronous speed. (ii) If the mechanical input is suddenly raised to 80 MW for an electrical load of 50 MW, find the rotor acceleration. (iii) If this acceleration is maintained for 10 cycles, find the change in torque angle and the rotor speed in rpm at the end of this period. *[Ref: Kothari & Nagrath, Ch-12, Example 12.1, p.435]*	[3]	CO4	C3	**TB**
	(c)	Two generators of 500 MVA (H = 4 MJ/MVA) and 300 MVA (H = 6 MJ/MVA) in the same station swing coherently. Find H_{eq} on a 100 MVA base. How do high-speed circuit breakers and auto-reclosing improve transient stability?	[3]	CO3	C4	**EXT**

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Semester Final Examination (OBE): **Final Model Question**

SET 2

Practice paper for self-study, mapped to CO1–CO5. Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

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Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

		Marks	CO	BL	Src
1.	(a) Define inductance. Show that the internal inductance of a conductor is ½ × 10^{-7} H/m. *[Ref: Kothari & Nagrath, Ch-2, Art 2.2–2.3, p.72]*	[3]	CO1	C3	**LN**
	(b) The outside diameter of the single layer of aluminium strands of an ACSR conductor is 5.04 cm. The diameter of each strand is 1.68 cm. Determine the 50 Hz reactance at 1 m spacing; neglect the effect of the central strand of steel and give reasons for doing so. *[Ref: Kothari & Nagrath, Ch-2, Example 2.2, p.81]*	[3]	CO1	C3	**TB**
	(c) Each conductor of the bundled-conductor line shown in Fig. 1 is ACSR, 1,272,000-cmil Pheasant (GMR = 0.0466 ft). Find the inductive reactance in Ω/km and Ω/mile per phase for d = 45 cm (f = 60 Hz).	[2]	CO1	C3	**PYQ**

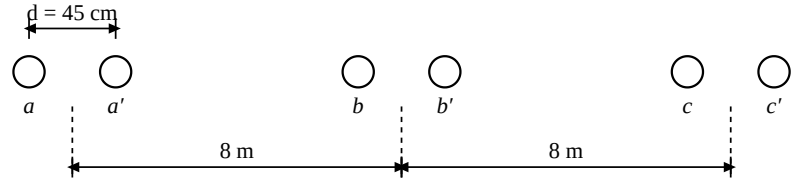


Fig. 1

[Ref: Asked 2021, 2022, 2024 (Stevenson, Example 4.5)]

	(d) Explain how lower series reactance from bundling increases the power-transfer capability of a line. Are there any drawbacks of bundling?	[2]	CO1	C4	**EXT**

2.	(a)	A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5 m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6 m between centres. The distance between the nearest conductors of the two lines is 20 m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150 A current flowing over the power line. *[Ref: Kothari & Nagrath, Ch-2, Problem 2.7, p.93]*	[3]	CO1	C3	**PYQ**
	(b)	Derive the capacitance of a single-phase two-wire line and show that the capacitance to neutral is twice the line-to-line capacitance. *[Ref: Kothari & Nagrath, Ch-3, Art 3.4, p.98]*	[3]	CO1	C3	**LN**
	(c)	A three-phase, 50 Hz overhead line has regularly transposed conductors equilaterally spaced 4 m apart. The capacitance of such a line is 0.01 μF/km. Recalculate the capacitance per km to neutral when the conductors are in the same horizontal plane with successive spacing of 4 m and are regularly transposed. *[Ref: Kothari & Nagrath, Ch-3, Problem 3.3, p.110]*	[2]	CO1	C3	**TB**
	(d)	Why must line capacitance be included in the model of a long EHV line? Explain what happens to the receiving-end voltage at no load.	[2]	CO1	C2	**EXT**
3.	(a)	Derive the expression for dielectric stress in a single-core cable and prove that it is maximum at the conductor surface. *[Ref: Asked 2021, 2022, 2024]*	[3]	CO2	C3	**PYQ**
	(b)	A single-core lead-sheathed cable has a conductor diameter of 3 cm; the diameter of the cable being 9 cm. The cable is graded by using two dielectrics of relative permittivity 5 and 4 respectively, with corresponding safe working stresses of 30 kV/cm and 20 kV/cm. Calculate the radial thickness of each insulation and the safe working voltage of the cable. *[Ref: V.K. Mehta, Ch-11, Example 11.13, p.283]*	[3]	CO2	C3	**TB**
	(c)	Find the economic size of a single-core cable working on a 220 kV, 3-phase system. The maximum permissible stress in the dielectric is not to exceed 250 kV/cm. *[Ref: V.K. Mehta, Ch-11, Tutorial Problem 2 (after Art 11.16), p.287]*	[2]	CO2	C3	**TB**
	(d)	A 400 kV AC submarine cable (C = 0.2 μF/km) is proposed for a 60 km sea crossing. Calculate its charging current and evaluate whether AC or HVDC should be chosen.	[2]	CO5	C5	**EXT**

OR

4.	(a)	Explain core-to-core (C_c) and core-to-earth (C_e) capacitances of a 3-core cable. Show that $C_N = C_e + 3C_c$ and explain how C_e and C_c are measured. *[Ref: V.K. Mehta, Ch-11, Art 11.14–11.15, p.288]*	[3]	CO2	C2	**LN**
	(b)	The capacitances of a 3-phase belted cable are 12.6 µF between the three cores bunched together and the lead sheath and 7.4 µF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. *[Ref: V.K. Mehta, Ch-11, Example 11.18, p.290]*	[2]	CO2	C3	**PYQ**
	(c)	The capacitance per kilometre of a 3-phase belted cable is 0.18 µF between two cores with the third core connected to sheath. Calculate the kVA taken by a 20 km long cable when connected to a 3-phase, 50 Hz, 3300 V supply. *[Ref: V.K. Mehta, Ch-11, Example 11.19, p.290]*	[3]	CO2	C3	**TB**
	(d)	Why are three single-core cables used instead of one 3-core cable for voltages above 66 kV?	[2]	CO5	C5	**EXT**

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL	Src
5.	(a)	Derive the expressions for sag when (i) the supports are at equal levels and (ii) the supports are at unequal levels. *[Ref: V.K. Mehta, Ch-8, Art 8.16, p.187]*	[3]	CO2	C3	**LN**
	(b)	A transmission line has a span of 200 m between level supports. The conductor has a cross-sectional area of 1.29 cm², weighs 1170 kg/km and has a breaking stress of 4218 kg/cm². Calculate the sag for a safety factor of 5, allowing a wind pressure of 122 kg per square metre of projected area. What is the vertical sag? *[Ref: V.K. Mehta, Ch-8, Example 8.19, p.191]*	[3]	CO2	C3	**TB**
	(c)	A 132 kV transmission line has the following data: weight of conductor = 680 kg/km, length of span = 260 m, ultimate strength = 3100 kg, safety factor = 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 m. *[Ref: V.K. Mehta, Ch-8, Example 8.17, p.190]*	[2]	CO2	C3	**PYQ**
	(d)	In a hilly region, towers often stand at different levels. Why does sag increase in summer, and what safety hazards arise if the ground clearance becomes insufficient?	[2]	CO4	C4	**EXT**

6.	(a)	Name the types of insulators with their voltage ranges. Why are suspension insulators preferred above 33 kV? *[Ref: V.K. Mehta, Ch-8, Art 8.5, p.164]*	[2]	CO2	C2	**LN**
	(b)	A string of four insulators has a self-capacitance equal to 5 times the pin-to-earth capacitance. Find (i) the voltage distribution across the various units expressed as a percentage of total voltage across the string and (ii) the string efficiency. *[Ref: V.K. Mehta, Ch-8, Example 8.9, p.177]*	[3]	CO2	C4	**TB**
	(c)	It is required to grade a string having seven suspension insulators. If the pin-to-earth capacitances are all equal to C, determine the line-to-pin capacitance that would give the same voltage across each insulator of the string. *[Ref: V.K. Mehta, Ch-8, Example 8.12, p.179]*	[3]	CO2	C4	**TB**
	(d)	In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11 % of the self-capacitance of each insulator, find (i) the distribution of voltage over the 3 insulators and (ii) the string efficiency. *[Ref: V.K. Mehta, Ch-8, Example 8.1, p.171]*	[2]	CO2	C3	**PYQ**

OR

7.	(a)	What is corona? Write down its advantages and disadvantages. *[Ref: Asked 2021, 2022, 2024]*	[3]	CO4	C2	**PYQ**
	(b)	Taking the dielectric strength of air to be 30 kV/cm, calculate the disruptive critical voltage for a 3-phase line with conductors of 1 cm radius and spaced symmetrically 4 m apart. *[Ref: V.K. Mehta, Ch-8, Tutorial Problem 2 (after Example 8.16), p.186]*	[2]	CO4	C3	**TB**
	(c)	A certain 3-phase equilateral transmission line has a total corona loss of 53 kW at 106 kV and a loss of 98 kW at 110.9 kV. What is the disruptive critical voltage? What is the corona loss at 113 kV? *[Ref: V.K. Mehta, Ch-8, Example 8.16, p.185]*	[3]	CO4	C4	**TB**
	(d)	A 220 kV line passes through a hilly area at high altitude where the air pressure is low. Analyse how the air density factor δ affects the critical disruptive voltage and the corona loss.	[2]	CO4	C4	**EXT**

8. (a) Using the P–δ curve, explain what happens for a sudden increase in mechanical input to a generator. [2] CO4 C2 **PYQ**

[Ref: Asked 2024 ×2, TT-3]

(b) In the system of Fig. 2, the generator has an inertia constant of 4 MJ/MVA ($E' = 1.075\angle 33.9°$ pu, P_{max} during a 3-phase fault at the middle of one line = 0.694 pu). Write the swing equation upon occurrence of the fault. What is the initial angular acceleration? If this acceleration can be assumed to remain constant for $\Delta t = 0.05$ s, find the rotor angle at the end of this time interval and the new acceleration. [3] CO4 C4 **TB**

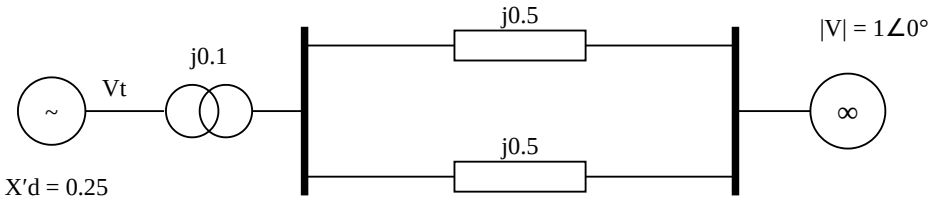


Fig. 2

[Ref: Kothari & Nagrath, Ch-12, Example 12.4, p.444]

(c) Find the steady-state power limit of a system consisting of a generator of equivalent reactance 0.50 pu connected to an infinite bus through a series reactance of 1.0 pu. The terminal voltage of the generator is held at 1.20 pu and the voltage of the infinite bus is 1.0 pu. Compare it with the limit if the generator emf is held fixed at 1.2 pu. [3] CO4 C4 **TB**

[Ref: Kothari & Nagrath, Ch-12, Example 12.6, p.447]

(d) Bangladesh imports power from India through an HVDC back-to-back station. Explain how an HVDC link (or a FACTS device) helps power transfer and stability between two AC grids. [2] CO3 C5 **EXT**

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SET 3

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Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

			Marks	CO	BL	Src
1.	(a)	Derive the expression for the flux linkages of one conductor in a group of conductors. *[Ref: Asked 2022]*	[3]	CO1	C3	**PYQ**
	(b)	A double-circuit three-phase line is shown in Fig. 1. The conductors a, a′; b, b′ and c, c′ belong to the same phase respectively. The radius of each conductor is 1.5 cm. Find the inductance of the double-circuit line in mH/km/phase.	[3]	CO1	C3	**TB**

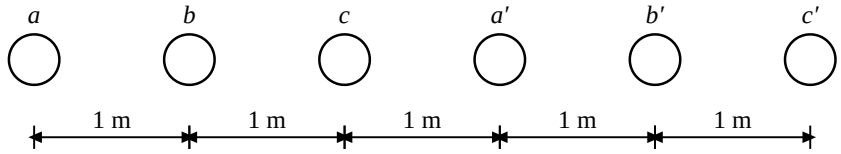


Fig. 1

[Ref: Kothari & Nagrath, Ch-2, Problem 2.12, p.94]

			Marks	CO	BL	Src
	(c)	A telephone line runs parallel to an untransposed three-phase transmission line, as shown in Fig. 2. The power line carries a balanced current of 400 A per phase. Find the mutual inductance between the circuits and calculate the 50 Hz voltage induced in the telephone line per km.	[2]	CO1	C4	**TB**



Fig. 2

[Ref: Kothari & Nagrath, Ch-2, Problem 2.8, p.93]

			Marks	CO	BL	Src
	(d)	A new 132 kV line will run parallel to a telecom cable route for 10 km. Recommend two measures to limit the induced voltage in the telecom circuit and justify them.	[2]	CO5	C5	**EXT**

2. (a) Define skin effect. Why is the effective AC resistance of a conductor higher than its DC resistance? [2] CO1 C2 **PYQ**

[Ref: Asked 2024]

(b) Derive the expression for the capacitance of a three-phase line with equilateral spacing. [2] CO1 C3 **LN**

[Ref: Kothari & Nagrath, Ch-3, Art 3.5, p.99]

(c) Consider the 500 kV, three-phase bundled-conductor line of Fig. 3. Find the capacitive reactance to neutral in Ω-km at 50 Hz. [2] CO1 C3 **TB**

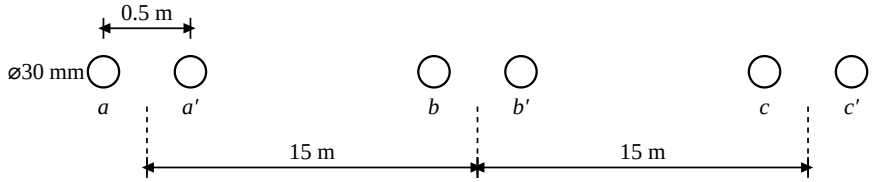


Fig. 3

[Ref: Kothari & Nagrath, Ch-3, Problem 3.4 (line of Fig. P-2.9), p.110]

(d) An overhead line 50 km in length is to be constructed of conductors 2.56 cm in diameter, for single-phase transmission. The line reactance must not exceed 31.4 Ω. Find the maximum permissible spacing. [2] CO1 C3 **TB**

[Ref: Kothari & Nagrath, Ch-2, Problem 2.10, p.94]

(e) A 132 kV, 50 Hz, fully transposed line has its conductors (diameter 1.8 cm) at the corners of a triangle with sides 4 m, 5 m and 6 m. Find the capacitance to neutral and the charging MVAr of a 200 km length. [2] CO1 C3 **EXT**

3. (a) Compare the merits and demerits of the underground system against the overhead system. Briefly describe the methods of laying underground cables. [3] CO2 C2 **LN**

[Ref: V.K. Mehta, Ch-11, Art 11.6 & Chapter Review, p.271]

(b) Calculate the capacitance and charging current of a single-core cable used on a 3-phase, 66 kV system. The cable is 1 km long, having a core diameter of 10 cm and an impregnated paper insulation of thickness 7 cm. The relative permittivity of the insulation may be taken as 4 and the supply at 50 Hz. [2] CO2 C3 **PYQ**

[Ref: V.K. Mehta, Ch-11, Example 11.5, p.276]

(c) A single-core cable of conductor diameter 2 cm and lead sheath of diameter 5.3 cm is to be used on a 66 kV, 3-phase system. Two intersheaths of diameter 3.1 cm and 4.2 cm are introduced between the core and lead sheath. If the maximum stress in the layers is the same, find the voltages on the intersheaths. [3] CO2 C4 **TB**

[Ref: V.K. Mehta, Ch-11, Example 11.15, p.285]

(d) Which cable type (belted, H-type or oil-filled) would you select for an 11 kV, a 33 kV and a 132 kV city link? Justify each choice. [2] CO5 C5 **EXT**

OR

4.	(a)	Define thermal resistance. Derive the thermal resistance of the dielectric of a single-core cable and the expression for the permissible current loading. *[Ref: V.K. Mehta, Ch-11, Art 11.17–11.19, p.291]*	[3]	CO2	C3	**LN**
	(b)	The capacitance per kilometre of a 3-phase belted cable is 0.3 μF between the two cores with the third core connected to the lead sheath. Calculate the charging current taken by five kilometres of this cable when connected to a 3-phase, 50 Hz, 11 kV supply. *[Ref: V.K. Mehta, Ch-11, Example 11.17, p.289]*	[2]	CO2	C3	**TB**
	(c)	The capacitances per kilometre of a 3-phase cable are 0.63 μF between the three cores bunched together and the sheath and 0.37 μF between one core and the other two connected to the sheath. Calculate the charging current taken by eight kilometres of this cable when connected to a 3-phase, 50 Hz, 6600 V supply. *[Ref: V.K. Mehta, Ch-11, Tutorial Problem 1 (after Example 11.19), p.290]*	[2]	CO2	C3	**TB**
	(d)	A single-core cable is laid in the ground, the core diameter being 30 mm and the dielectric thickness 40 mm. The thermal resistivity of the dielectric is 5 thermal ohm-metres and the thermal resistance between the sheath and the ground surface is 0.45 thermal ohm per metre length. Neglecting dielectric and sheath losses, estimate the maximum permissible current loading if the temperature difference between the conductor and the ground surface is not to exceed 55 °C. The electrical resistance of the cable is 110 μΩ per metre. *[Ref: V.K. Mehta, Ch-11, Example 11.20, p.292 (also in your TT-2)]*	[3]	CO2	C3	**TB**

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL	Src
5.	(a)	What is string efficiency? Discuss the methods of improving string efficiency. *[Ref: Asked 2021, 2022, 2024]*	[3]	CO2	C2	**PYQ**
	(b)	An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency. *[Ref: V.K. Mehta, Ch-8, Example 8.5, p.173]*	[2]	CO2	C4	**TB**
	(c)	A string of 4 insulators has a self-capacitance equal to 10 times the pin-to-earth capacitance. Find (i) the voltage distribution across various units expressed as a percentage of total voltage across the string and (ii) the string efficiency. *[Ref: V.K. Mehta, Ch-8, Example 8.6, p.174]*	[3]	CO2	C4	**TB**
	(d)	A 132 kV line string has an efficiency of only 76 %. Evaluate whether adding one more disc or fitting a guard ring is the better remedy.	[2]	CO5	C5	**EXT**

6. (a) Explain the effect of wind and ice loading on sag. Write the expressions for slant sag and vertical sag. [2] CO2 C2 **LN**

[Ref: V.K. Mehta, Ch-8, Art 8.16, p.189]

(b) An overhead transmission line conductor having a parabolic configuration weighs 1.925 kg per metre of length. The area of X-section of the conductor is 2.2 cm^2 and the ultimate strength is 8000 kg/cm^2. The supports are 600 m apart, having 15 m difference of levels. Calculate the sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5. Assume that the ice load is 1 kg per metre run and there is no wind pressure. [3] CO2 C3 **TB**

[Ref: V.K. Mehta, Ch-8, Example 8.24, p.194]

(c) An overhead line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm^2. The ultimate strength is 5000 kg/cm^2 and the safety factor is 5. The specific gravity of the material is 8.9 gm/cc. The wind pressure is 1.5 kg/m. Calculate the height of the conductor above the ground level at which it should be supported if a minimum clearance of 7 m is to be left between the ground and the conductor. [3] CO2 C3 **TB**

[Ref: V.K. Mehta, Ch-8, Example 8.22, p.192]

(d) An overhead transmission line at a river crossing is supported from two towers at heights of 40 m and 90 m above water level, the horizontal distance between the towers being 400 m. If the maximum allowable tension is 2000 kg, find the clearance between the conductor and water at a point mid-way between the towers. Weight of conductor is 1 kg/m. [2] CO2 C3 **PYQ**

[Ref: V.K. Mehta, Ch-8, Example 8.25, p.195]

OR

7. (a) With a neat diagram, explain the Varley loop test for locating an earth fault and derive the fault-distance expression. [3] CO4 C3 **LN**

[Ref: V.K. Mehta, Ch-11, Art 11.23, p.295]

(b) Murray loop test is performed on a faulty cable 300 m long. At balance, the resistance connected to the faulty core was set at 15 Ω and the resistance of the resistor connected to the sound core was 45 Ω. Calculate the distance of the fault point from the test end. [2] CO4 C3 **TB**

[Ref: V.K. Mehta, Ch-11, Example 11.21, p.296]

(c) The Varley loop test is used to find the position of an earth fault on a line of length 40 km. The resistance/km of a single line is 28 Ω. The fixed resistors have resistances of 250 Ω each. The fault is calculated to be 7 km from the test end. To what value was the variable resistor set? [2] CO4 C4 **TB**

[Ref: V.K. Mehta, Ch-11, Tutorial Problem 3 (after Example 11.24), p.298]

(d) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. [3] CO4 C3 **PYQ**

[Ref: V.K. Mehta, Ch-11, Example 11.23, p.297]

8. (a) Define HVDC. Discuss the economics of power transmission with a cost–distance curve. [3] CO5 C2 **PYQ**

[Ref: Asked 2022, 2024]

(b) Explain series and shunt compensation of transmission lines. Name the main FACTS controllers. [2] CO3 C2 **LN**

[Ref: Lecture 16–17]

(c) A generator (X_{dg} = 1 pu) supplies a motor (X_{dm} = 1 pu) through a transformer (0.1 pu), a line (0.25 pu) and a transformer (0.1 pu). A three-phase static capacitive reactor of reactance 1 pu per phase is connected through a switch at the motor bus bar. Calculate the limit of steady-state power with and without the reactor switch closed. Recalculate the power limit with the capacitive reactor replaced by an inductive reactor of the same value. Assume the internal voltage of the generator to be 1.2 pu and that of the motor to be 1.0 pu. [3] CO4 C4 **TB**

[Ref: Kothari & Nagrath, Ch-12, Example 12.2, p.440]

(d) A lossless line has X = 0.8 pu with $|V_s| = |V_r| = 1.0$ pu. Find the maximum power transfer before and after 40 % series compensation, and comment on the use of TCSC for this purpose. [2] CO3 C4 **EXT**

How These Sets Are Built

Pattern assumed (as you described): about **20 %** previous-year questions; about **60 %** from the textbook (worked examples, end-of-chapter exercises) and the course instructor's lecture notes; about **20–25 %** added or modified by the external examiner (application, "justify / compare / evaluate" questions mapped to the COs).

Set	CO1	CO2	CO3	CO4	CO5	PYQ	Textbook + lecture	External style
Set 1	16	28	3	25	8	13–14 (22–23 %)	32–34 (53–57 %)	13–14 (22–23 %)
Set 2	20	34	2	20	4	13–15 (22–25 %)	33–37 (55–62 %)	10–12 (17–20 %)
Set 3	18	36	4	13	9	13–16 (22–27 %)	34–39 (57–65 %)	8–10 (13–17 %)

CO columns are printed marks out of 80. Source columns are out of the 60 marks you answer; the range depends on which optional questions (Q3/Q4, Q6/Q7) you choose.

Books referred to

- **Kothari & Nagrath**, *Modern Power System Analysis*: Ch-2 Inductance & Resistance, Ch-3 Capacitance, Ch-12 Power System Stability (01_Textbook_Chapters/Chap_1_and_2.pdf, Chap_12_Nagarath_Stability.pdf)
- **V.K. Mehta**, *Principles of Power System*: Ch-8 Mechanical Design of Overhead Lines, Ch-11 Underground Cables (01_Textbook_Chapters/Chap_8_Mehta.pdf, Chap_11_Mehta.pdf)

Every textbook maths problem in the three sets

- **Set 1:** 1(c) Kothari & Nagrath, Ch-2, Example 2.3, p.81; 2(b) Kothari & Nagrath, Ch-3, Problem 3.8, p.111; 2(c) Kothari & Nagrath, Ch-3, Example 3.1, p.103; 3(b) V.K. Mehta, Ch-11, Example 11.1, p.273; 3(c) V.K. Mehta, Ch-11, Example 11.9, p.279; 4(b) V.K. Mehta, Ch-11, Example 11.12, p.282; 4(c) V.K. Mehta, Ch-11, Example 11.2, p.274; 5(b) V.K. Mehta, Ch-8, Example 8.8, p.176; 5(c) V.K. Mehta, Ch-8, Example 8.11, p.179; 6(b) V.K. Mehta, Ch-8, Example 8.13, p.184; 6(c) V.K. Mehta, Ch-8, Example 8.15, p.185; 7(c) V.K. Mehta, Ch-11, Example 11.22, p.297; 7(d) V.K. Mehta, Ch-11, Example 11.24, p.297; 8(b) Kothari & Nagrath, Ch-12, Example 12.1, p.435
- **Set 2:** 1(b) Kothari & Nagrath, Ch-2, Example 2.2, p.81; 2(a) Kothari & Nagrath, Ch-2, Problem 2.7, p.93; 2(c) Kothari & Nagrath, Ch-3, Problem 3.3, p.110; 3(b) V.K. Mehta, Ch-11, Example 11.13, p.283; 3(c) V.K. Mehta, Ch-11, Tutorial Problem 2 (after Art 11.16), p.287; 4(b) V.K. Mehta, Ch-11, Example 11.18, p.290; 4(c) V.K. Mehta, Ch-11, Example 11.19, p.290; 5(b) V.K. Mehta, Ch-8, Example 8.19, p.191; 5(c) V.K. Mehta, Ch-8, Example 8.17, p.190; 6(b) V.K. Mehta, Ch-8, Example 8.9, p.177; 6(c) V.K. Mehta, Ch-8, Example 8.12, p.179; 6(d) V.K. Mehta, Ch-8, Example 8.1, p.171; 7(b) V.K. Mehta, Ch-8, Tutorial Problem 2 (after Example 8.16), p.186; 7(c) V.K. Mehta, Ch-8, Example 8.16, p.185; 8(b) Kothari & Nagrath, Ch-12, Example 12.4, p.444; 8(c) Kothari & Nagrath, Ch-12, Example 12.6, p.447
- **Set 3:** 1(b) Kothari & Nagrath, Ch-2, Problem 2.12, p.94; 1(c) Kothari & Nagrath, Ch-2, Problem 2.8, p.93; 2(c) Kothari & Nagrath, Ch-3, Problem 3.4 (line of Fig. P-2.9), p.110; 2(d) Kothari & Nagrath, Ch-2, Problem 2.10, p.94; 3(b) V.K. Mehta, Ch-11, Example 11.5, p.276; 3(c) V.K. Mehta, Ch-11, Example 11.15, p.285; 4(b) V.K. Mehta, Ch-11, Example 11.17, p.289; 4(c) V.K. Mehta, Ch-11, Tutorial Problem 1 (after Example 11.19), p.290; 4(d) V.K. Mehta, Ch-11, Example 11.20, p.292 (also in your TT-2); 5(b) V.K. Mehta, Ch-8, Example 8.5, p.173; 5(c) V.K. Mehta, Ch-8, Example 8.6, p.174; 6(b) V.K. Mehta, Ch-8, Example 8.24, p.194; 6(c) V.K. Mehta, Ch-8, Example 8.22, p.192; 6(d) V.K. Mehta, Ch-8, Example 8.25, p.195; 7(b) V.K. Mehta, Ch-11, Example 11.21, p.296; 7(c) V.K. Mehta, Ch-11, Tutorial Problem 3 (after Example 11.24), p.298; 7(d) V.K. Mehta, Ch-11, Example 11.23, p.297; 8(c) Kothari & Nagrath, Ch-12, Example 12.2, p.440

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